

Functional Requirements Document (FRD)

Suraksha Shadow – Women Safety & Social Impact

Version: 1.0 · **Prepared for:** Hackathon Round 1
Submission

1. Purpose

This document defines the functional requirements for Suraksha Shadow, an AI-powered personal safety companion. It describes *what* the system must do, independent of *how* it is built (see the companion Technical Requirements Document for implementation details).

2. Project Overview

Suraksha Shadow pairs silent, real-time protection with structured after-incident support, built around three modules:

- **Shield** (Phase 1) – silently detects distress and alerts trusted contacts without tipping off a threat
- **Guardian Mode** (Phase 1) – gives trusted contacts a live situational dashboard once Shield fires
- **Sahara** (Phase 2) – a trauma-informed AI companion that guides the user through next steps after an incident

3. Scope

In scope – Phase 1 (this hackathon): Shield's detection, silent SOS, fake-call deception, and Guardian Mode's live dashboard.

In scope – Phase 2 (roadmap): Sahara's guided support chat and knowledge base.

Out of scope (all phases): direct integration with police/emergency dispatch systems, hardware/wearable devices, automated law-enforcement reporting, multi-language support beyond English/Hindi.

4. User Roles

Role	Description
Primary User	The person using Suraksha Shadow for protection

Role	Description
Trusted Contact (Guardian)	A person the Primary User has designated to receive alerts and Guardian Mode data
System	The application's backend and AI components, acting on the Primary User's behalf

5. Functional Requirements

5.1 Shield (Phase 1)

ID	Requirement	Description
FR-1	Silent Trigger Detection	The system shall detect a distress signal via a spoken code word (during a real or simulated call) or a sudden motion anomaly (e.g., running, struggling), without requiring a visible, deliberate action from the Primary User.
FR-2	Silent SOS Dispatch	On trigger, the system shall send the Primary User's live location and a short audio clip to all designated

ID	Requirement	Description
		Trusted Contacts (target: within 3 seconds), with no visible on-screen alert.
FR-3	Fake-Call Deception	Upon triggering, the system shall generate a natural-sounding simulated conversation (audio) so it appears the Primary User is engaged in an ordinary phone call.

5.2 Guardian Mode (Phase 1)

ID	Requirement	Description
FR-4	Live Location Sharing	Once Shield fires, the system shall continuously share the Primary User's location with Trusted Contacts until the emergency is resolved or manually ended.
FR-5	Consent-Gated Ambient Audio	The system shall stream ambient audio to Trusted Contacts only if the Primary User has explicitly

ID	Requirement	Description
		granted consent in advance.
FR-6	Consent Management	The Primary User shall be able to grant or revoke ambient-audio consent at any time, independent of an active emergency.
FR-7	Battery Status Reporting	The system shall report the Primary User's device battery percentage to Trusted Contacts for the duration of the emergency.
FR-8	Movement Status Detection	The system shall report whether the Primary User's device is stationary or in motion, and the nature of that motion, to Trusted Contacts.
FR-9	Emergency Timeline Log	The system shall maintain a timestamped log of key events during an emergency (trigger fired, location updates, contact acknowledgment) and make it visible to Trusted Contacts.

5.3 Sahara (Phase 2)

ID	Requirement	Description
FR-10	Trauma-Informed AI Chat	The system shall automatically open a calm, plain-language conversational interface for the Primary User immediately after a Shield SOS fires.
FR-11	Guided Next-Steps Flow	The system shall guide the Primary User through filing a complaint, documenting evidence, and locating the nearest relevant help desk.
FR-12	Knowledge-Base-Grounded Answers	The system shall answer user questions using retrieval over a curated knowledge base (relevant BNS sections, the POSH Act, and verified NGO/helpline directories) rather than unconstrained generation.

6. Key User Stories

- As a **Primary User**, I want help to be summoned without anyone around me noticing, so that I'm not put at greater risk by asking for help.
- As a **Trusted Contact**, I want to see more than just a location pin during an emergency, so that I can judge how serious the situation is and respond appropriately.
- As a **Primary User**, I want clear control over whether my ambient audio is shared, so that I trust the app isn't recording me without my say-so.
- As a **Primary User**, I want guidance on what to do after an incident, so that I'm not left figuring out complaint procedures and evidence steps alone.

7. Assumptions & Constraints

- Primary Users grant location, microphone, and motion-sensor permissions during setup.
- At least one Trusted Contact must be configured before Shield or Guardian Mode can function.
- Sahara's guidance is India-specific (BNS, POSH Act) for this version.
- The system requires an active internet connection; full offline operation is out of scope for Phase 1.